

A preliminary note on the landslide disaster of 26.08.2025 at Ardhkuwari on the Vaishno Devi Yatra Route, Katra, Reasi District, UT: Jammu & Kashmir

Geological Survey of India

The UT: Jammu and Kashmir has been experiencing heavy to extreme amount of rainfall in recent times, triggering flash floods and landslide hazards that have been disrupting communication/ Yatra routes, damaged vital infrastructure, and caused widespread damages, and loss of precious lives. On 26.08.2025, a major landslide disaster occurred along the Yatra route from Banganga to Ardhkuwari, near the Inderprastha Bhojnalaya at Ardhkuwari (Fig-1), at around 3:00 PM. The landslide hazards had huge impact and resulted in 32 casualties and left at least 20 persons injured, as per the reports of the Shri Mata Vaishno Devi Shrine Board (SMVDSB) and media sources (Fig-2 a, b, c & d). In view of safety concerns, the Yatra has been temporarily suspended by Shri Mata Vaishno Devi Shrine Board (SMVDSB) until weather conditions improve.

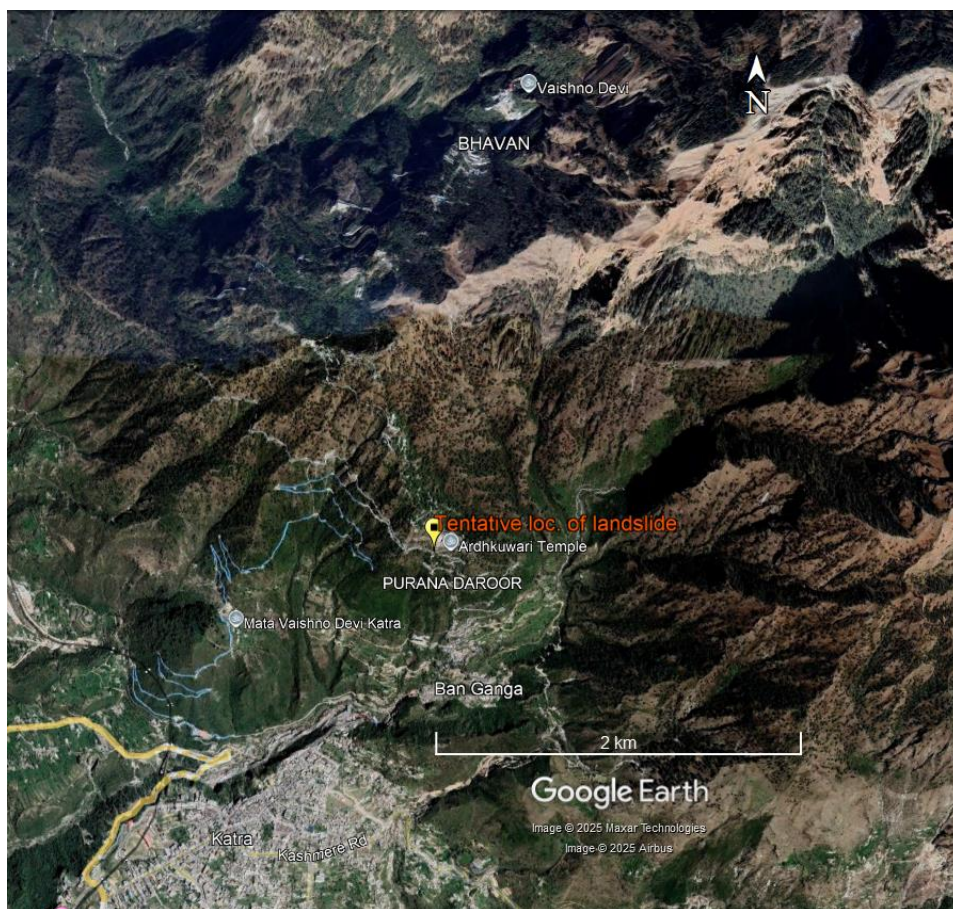
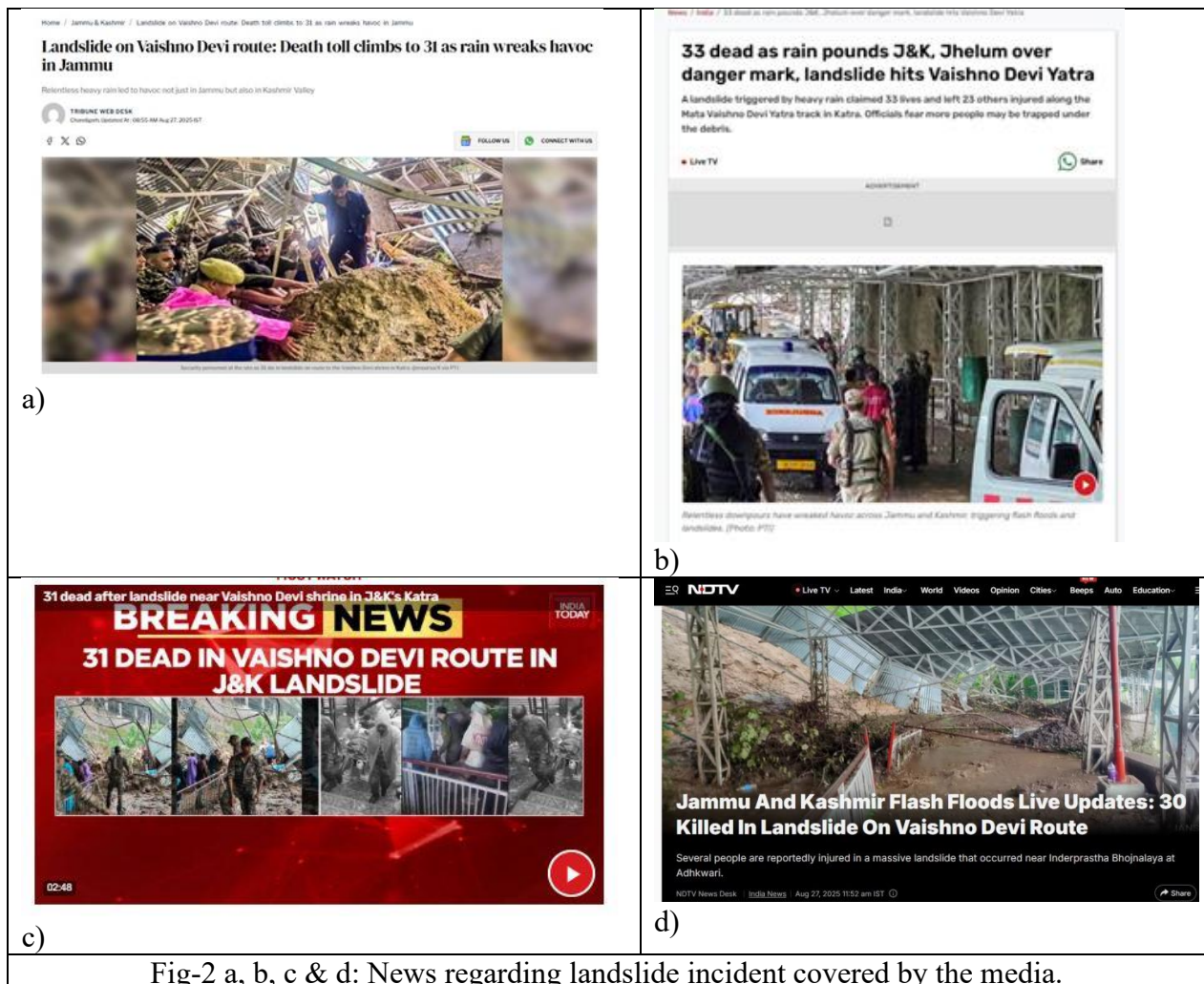


Fig-1: Google image showing the tentative location of the landslide near Ardhkuwari



The yatra tracks from Katra to Bhawan were constructed by cutting through steep slopes. Many incidences of landslides along these tracks have been reported in the past, especially in the rainy season every year. Shooting stones are a frequent occurrence, and there are several vulnerable spots with slope instability issues along the routes.

Geologically the area is represented by limestone, dolomite, and stromatolitic dolomite of the Trikuta Formation of the Sirban Group. Moreover, scree deposits of Quaternary age belonging to the Vaishnodevi Formation are exposed to the south of this affected area. These scree deposits vary from older, well-compacted material to older, poorly compacted material to loose material. Reasi Thrust, represented by the juxtaposition of the Sirban Group of rocks with the Murree Group of rocks, is the major regional tectonic discontinuity present in proximity of the area and the same is trending in a NW-SE direction. (Fig-3).

The Geological Survey of India (GSI) has already carried out geoscientific studies on various scales in this particular study area, and have highlighted its vulnerability to landslide hazards in all its previous



scientific reports, which have already been submitted to the local authorities, and the J&K Government. The following is the list of the recent GSI studies carried out in the above affected area.

1. The affected area around Ardhkuwari was identified as a high-hazard zone in the study “Landslide Mesozonation of Vaishno Devi Area with Special Reference to Slope Stability.” [Kumar P, 2006-2007] (Fig-4).
2. In the “Macro-scale (1:50,000) Landslide Susceptibility Mapping in parts of Toposheet nos. 43K/16 and 43L/13, Jammu & Reasi districts, Jammu and Kashmir (J&K)”, the track route was categorized under high landslide susceptibility zones. [Bhat A A & Akaram V, 2016-17] (Fig-5).
3. During the “Site Specific Detailed Geological Mapping and Stability Analysis of Landslide near Siar Baba Waterfall area and Assessment of the Stability of Vulnerable Slope Stretches along the three tracks joining Katra with Shri Mata Vaishno Devi Shrine, Reasi District, Jammu and Kashmir”, several vulnerable slopes were identified. Notably, the slopes around the recent landslide incident had already been classified as vulnerable and marked as Nos. 110, 111, 115, and 117 in the map (Fig. 6, Annexure-1). Recommendations were made for further investigations and site-specific remedial measures. [Kumar P & Thapa D R, 2022-23].
4. The reports of the studies as mentioned above have already been handed over to SMVDSB for implementation which included suggested remedial measures (Annexure-1). The report pertaining to Siar Baba and the vulnerable zones along the Shri Mata Vaishno Devi Yatra track route was formally handed over to the Chief Executive Officer, Shri Mata Vaishno Devi Shrine Board, on 26.02.2024. Further, the NLSM report was submitted to the State Government on 20.05.2024.

Moreover, a tripartite Memorandum of Understanding (MoU) has been signed recently on 7th March, 2025, amongst Geological Survey of India, Shri Mata Vaishno Devi Shrine Board, Katra and THDC India Limited (THDCIL) to address slope instability along the entire Yatra route with the following objectives:

1. To identify and delineate areas of slope instability along the entire Yatra route, as per the requirements of the SMVDSB.
2. To suggest treatment measures for vulnerable zones.

Under the provisions of the above MoU, the Geological Survey of India will carry out slope stability studies along the Yatra route. The outcome of this collaborative effort is expected to facilitate the local authorities significantly to mitigate the landslide hazards in the Yatra route, and will enhance the safety and security of pilgrims, contributing to a safer and more secure journey to the sacred Shri Mata Vaishno Devi Shrine.

A team of GSI geoscientists is already at site of the landslide-affected area near Ardhkuwari to conduct a post disaster landslide assessment study of the incident of 26.08.2025. A preliminary report with the findings will be prepared thereafter, and will be submitted to the concerned authorities shortly.

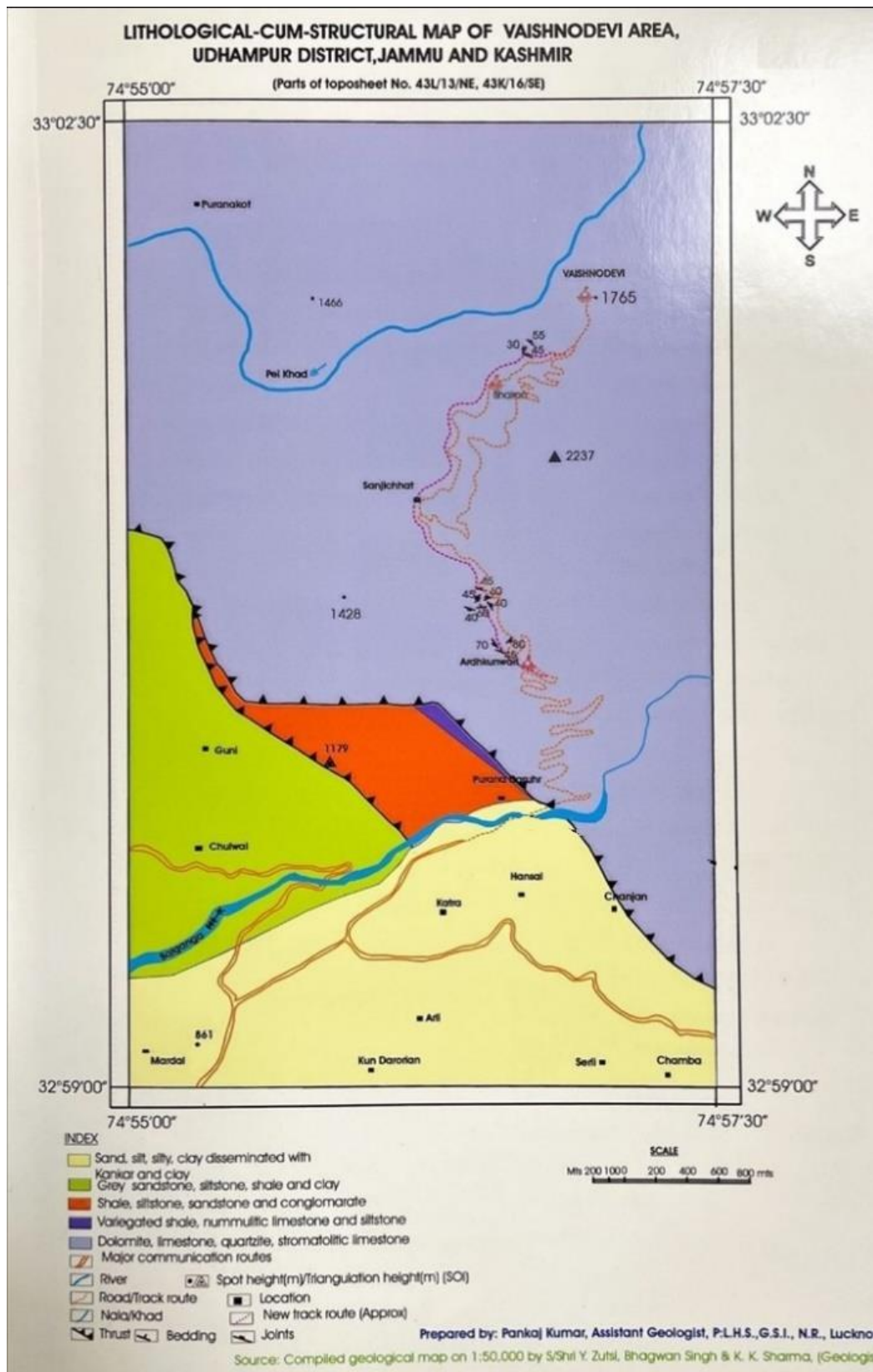


Fig-3: Lithological Cum Structural Map of the area

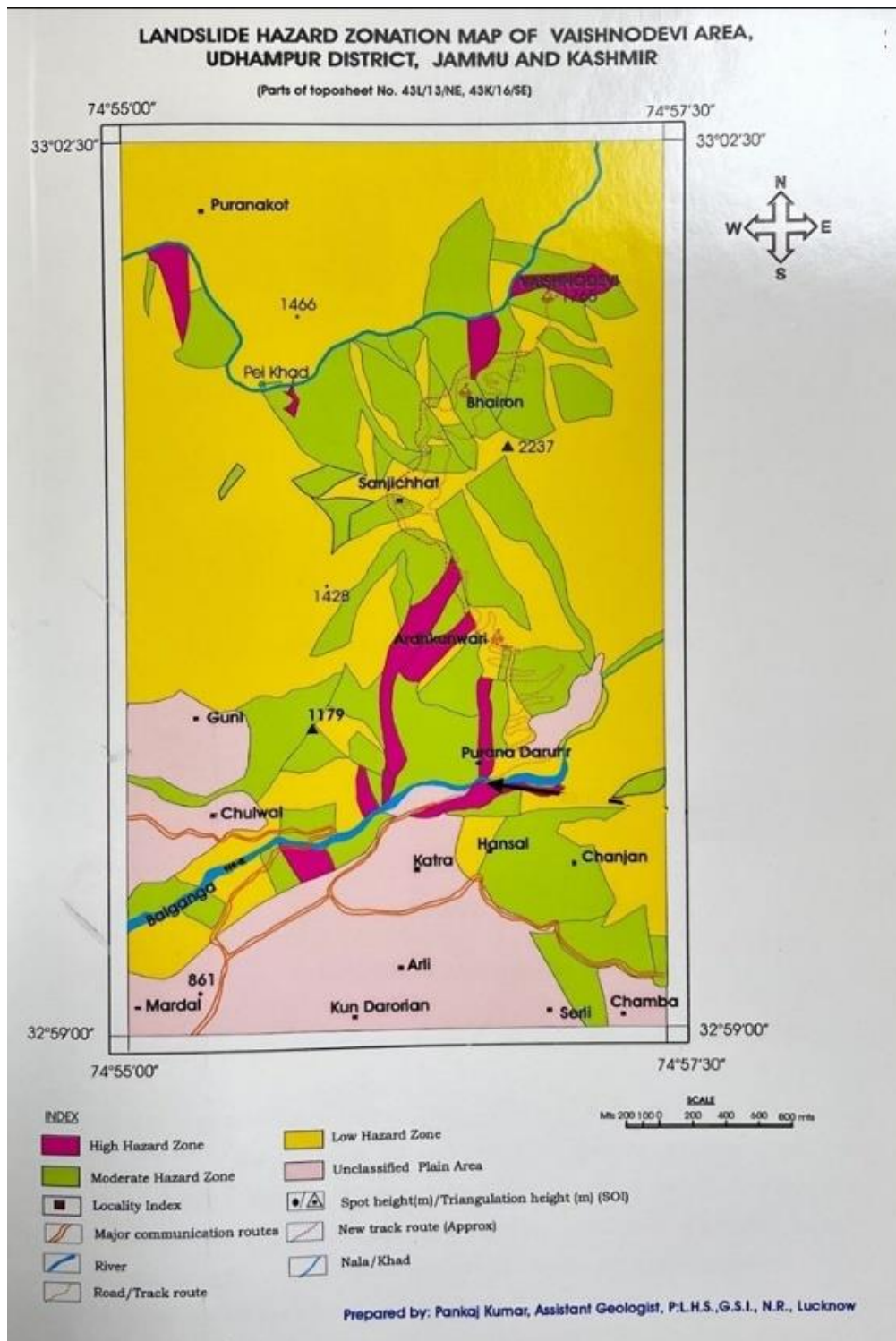
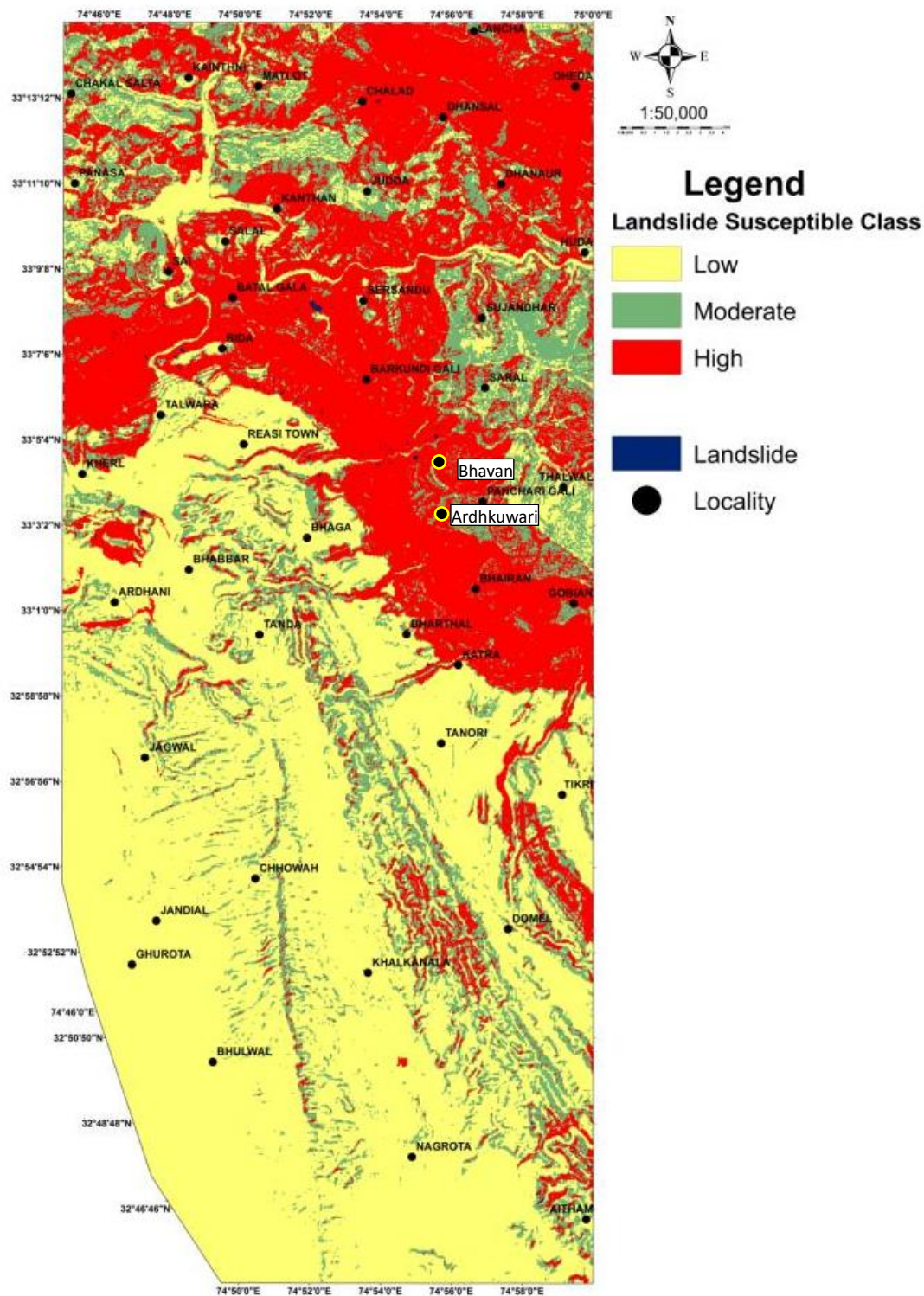


Fig-4: Landslide Hazard Zonation Map of Landslide area



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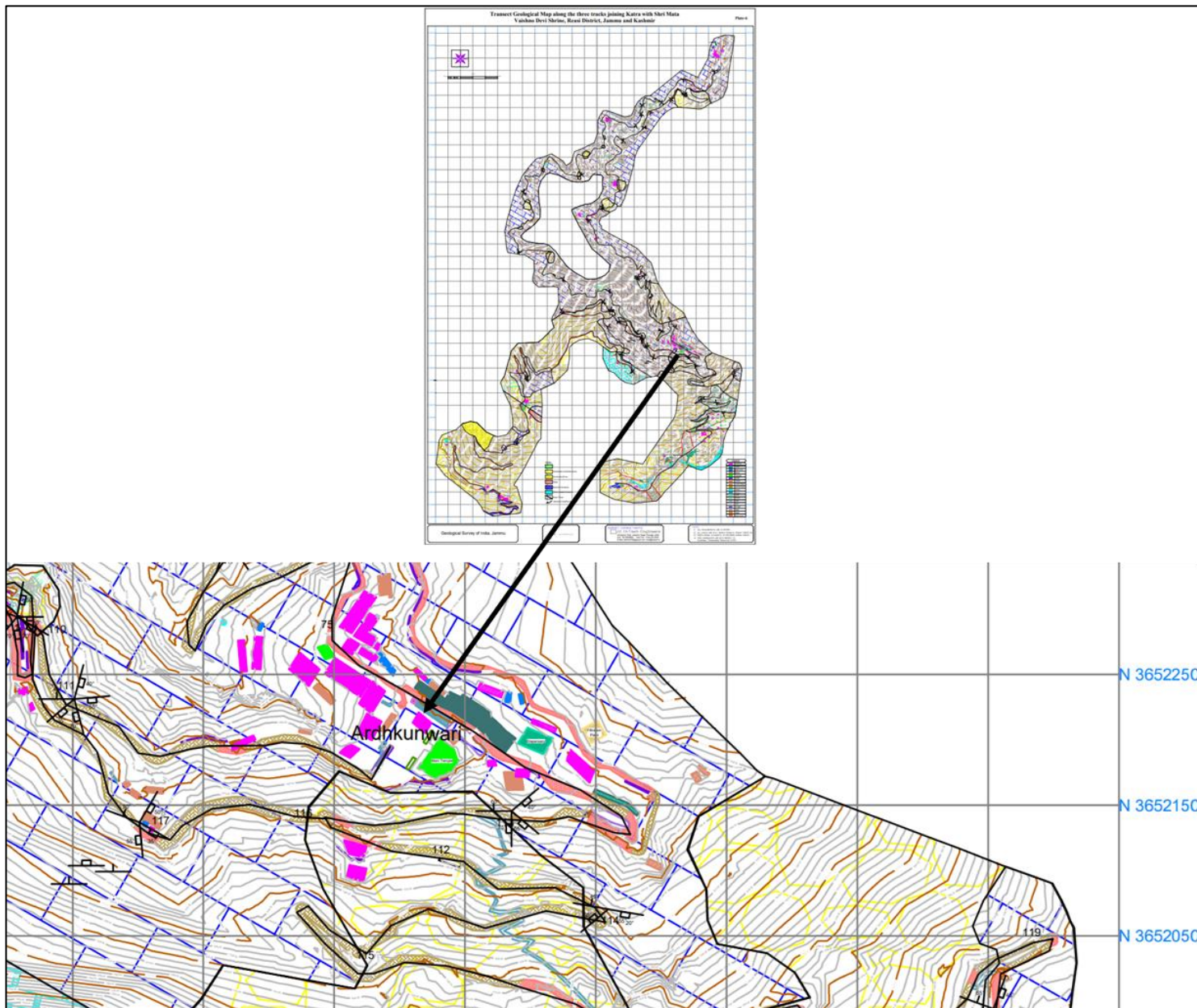


Fig-6: Transect Geological Map (1:10000 scale) along the three tracks joining Katra with Shri Mata Vaishno Devi Shrine, Reasi District, Jammu and Kashmir



Annexure - I

Table-7.1: Proposed remedial measures along all tracks									
Tarakot-Adhkuwari track									
Sl. No.	Loc.	Lat.			Long.			Probable Type of Failure	Remedial measures
1	2	33°	0'	33"	74°	56'	29.5"	Debris	Retaining wall with weep hole
2	3	33°	0'	35.7"	74°	56'	25.3"	Cut face	Retaining wall with sufficient height. Removal of loose blocks lying on slope.
3	7	33°	0'	39.4"	74°	56'	19.9"	Rockfall zone	Rockbolts/SFRS
4	9	33°	0'	41.4"	74°	56'	17.9"	Slope vulnerable for rock slide	Benching and toe wall
5	12	33°	0'	39.3"	74°	56'	5.2"	Rock slide	Rockbolts/SFRS
6	15	33°	0'	38"	74°	56'	17.4"	Cut face on left bank of nala	Removal of loose boulders and construction of breast wall with weep holes.
7	22	33°	0'	24.2"	74°	56'	26.4"	Rockfall zone	Hanging boulders should be removed, shotcrete with wiremesh net.
8	41	33°	0'	2.8"	74°	55'	31.5"	Detached boulder prone to slide (dimension 3x3.5 m approx.)	Remove or to be supported.
9	34 to 35	33°	0'	17.9"	74°	55'	52.9"	Rockfall zone	Removal of loose/hanging rock blocks and Fibre rich shotcrete. Bio-restoration of the slope to check the surface erosion near location no. 34 below the track.
10	35A	33°	0'	15.28"	74°	55'	53.74"	Cracks and Subsidence	(Annexure-III)
11	46	33°	0'	1.2"	74°	55'	40.5"	Detached boulder prone to slide (dimension 4x3 m approx.)	Remove or to be supported
12	48 to 48 A	33°	0'	3.7"	74°	55'	40.5"	Open cracks in the compacted scree making the slope forming material vulnerable to fall.	Removal of loose blocks and provide protection wall.
13	49A	33°	0'	2.8"	74°	55'	46.1"	Cut face/rock slide	SFRS and Breast wall with drainage hole



Sl. No.	Loc.	Lat.			Long.			Probable Type of Failure	Remedial measures
Bhawan-Sanjichat-Adhkuwari									
14	52	33°	1'	40.5"	74°	56'	42.9"	Vulnerable to planar and wedge failure	Dental treatment for shear zone 80°/N110°. Rock bolting.
15	55	33°	1'	36.3"	74°	56'	43.4"	Planar along some joint and many wedges	Scaling and shotcrete along the slope face
16	57	33°	1'	35.3"	74°	56'	38.6"	Planar failure	Scaling of debris and Rockbolts
17	59	33°	1'	29.3"	74°	56'	29.8"	Old planar failure	Scaling of debris material along slope and Rockbolts
18	60	33°	1'	22.2"	74°	56'	22.6"	Rockfall/shooting stone area	Scaling of slope and provide drapery.
19	62	33°	1'	15.8"	74°	56'	17.8"	Shooting stone zone	Scaling of slope/wiremesh
20	65	33°	1'	6"	74°	56'	16.1"	Topple /wedge	Benching
21	70	33°	0'	48.3"	74°	56'	28.8"	Planar/wedge	Support system already provided by the Shrine board
Bhawan-Adhkuwari (Himkoti Marg)									
22	77	33°	1'	33.9"	74°	56'	33.8"	Old planar failure	Rock bolting
23	78	33°	1'	35.1"	74°	56'	33.1"	Old planar failure	Rock bolting
24	80	33°	1'	35.6"	74°	56'	31.3"	Wedge failure	Rockfall barrier already provided
25	81	33°	1'	30.9"	74°	56'	27.9"	Shooting stone zone	Rockbolt with wiremesh and retaining wall
26	82	33°	1'	27.3"	74°	56'	25.8"	Shooting stone zone	Rockbolt with wiremesh and retaining wall
27	87	33°	1'	15.4"	74°	56'	5.8"	Planar	Rockbolt with wiremesh and retaining wall
28	91	33°	1'	10.5"	74°	55'	52.8"	Planar/Wedge	Shortcrete and toe wall
28	99	33°	0'	51.8	74°	56'	0.22"	Overhang	Trimming of Overhang/construction breast wall up to matching height.
29	104	33°	0'	44.6"	74°	56'	18.7"	Planar	Rockbolt and SFRS
30	106	33°	0'	40.7"	74°	56'	22.3"	Wedge failure	Remedial measures (Annexure-III)
31	110	33°	0'	32.3"	74°	56'	30.1"	Rockfall	Rockfall barrier



Sl. No.	Loc.	Lat.			Long.			Probable Type of Failure	Remedial measures
32	111	33°	0'	30.8"	74°	56'	30.8"	Planar/Wedge	Drapery
Banganga-Adhkuwari track									
33	115	33°	0'	24.2"	74°	56'	39.6"	Planar	SFRS
34	117	33°	0'	27.7"	74°	56'	34.3"	Planar/Wedge (Highly shattered rockmass)	SFRS
35	118	33°	0'	20.2"	74°	56'	51.5"	Shooting stone area	Rockfall barrier
36	119	33°	0'	24.1"	74°	56'	59"	Rockfall	Wiremesh net with shotcrete
37	120	33°	0'	17.4"	74°	56'	51.7"	Rock slide vulnerable zone	Toe wall with drainage holes
38	121	32°	59'	59.5"	74°	56'	47.4"	Debris slide	Remedial measures (Annexure-IV)
39	122	32°	59'	53.3"	74°	56'	32.4"	Debris slide	Retaining wall with drainage hole, bio-restoration.